Majorana Modes in the Machine

Final Synthesis: From the Kitaev Chain to a Noisy Quantum Processor

Dobromir Stoev
Edgar Harutyunyan
Guilherme Schewtschik
Jaskaran Singh
Zhengyi Liu
Zhenming Shi

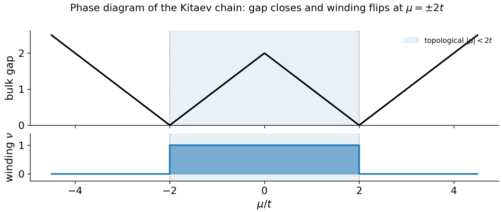
Advanced Seminar – AI-Augmented Theoretical Physics

July 4, 2026

The Question Behind the Project

Majorana zero modes are the textbook route to a topological qubit: information stored *non-locally* across the two ends of a wire.

- The **Kitaev chain** is the minimal model that hosts them.
- A real quantum processor is **noisy** (NISQ): finite T_1/T_2, imperfect gates, no error correction.



The topological window $|\mu| < 2t$ we will track throughout.

Central question

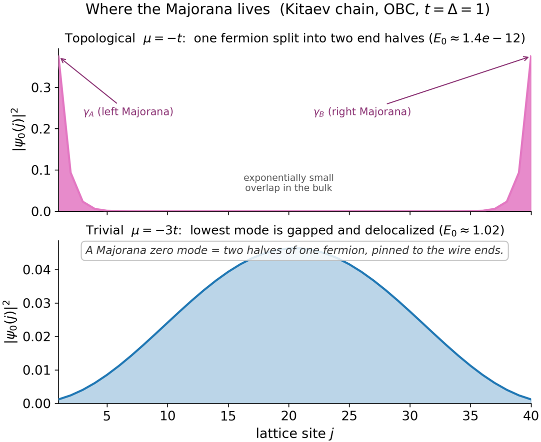
If we *simulate* the Kitaev chain on a gate-based quantum computer, does the topological signature survive the machine that runs it?

The Arc in Four Blocks

- **Block 1 – The physics.** Kitaev Hamiltonian, Majorana edge modes, the topological phase transition and its bulk invariant.
- **Block 2 – Onto qubits.** Jordan–Wigner mapping, fermion parity, and a *non-local* observable that sees the topology.
- **Block 3 – State preparation.** A parity-controlled VQE prepares the ground state across the phase diagram; the diagnostic is validated.
- **Block 4 – NISQ reality.** Circuit-level gate noise: what breaks the signal, what protects it, and how deep a circuit should be.

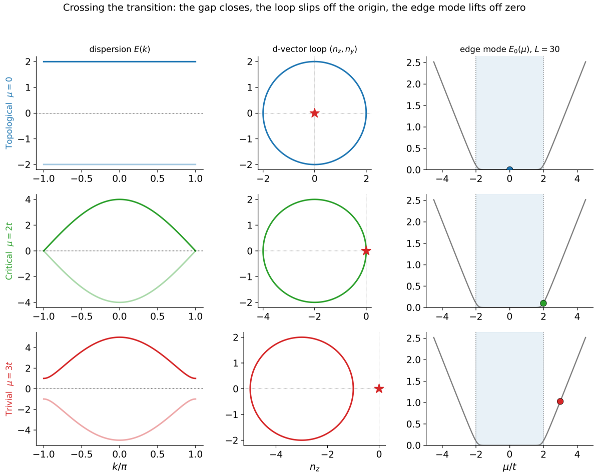
This talk follows one thread end-to-end: the **non-local edge string** $\langle X_0 Z_1 \cdots Z_{L-2} X_{L-1} \rangle$, from exact physics to a noisy circuit.

Block 1 – Where the Majorana Lives



In the **topological** phase ($|\mu| < 2t$) one fermion splits into two Majorana halves γ_A, γ_B pinned to the wire ends, at zero energy. In the **trivial** phase the lowest mode is gapped and delocalized. The two ends share *one* non-local fermion – that is the qubit.

Block 1 – A Genuine Phase Transition



The bulk gap closes exactly at $|\mu| = 2t$ and reopens: the two phases cannot be connected without a gap closing.

- Topological ($|\mu| < 2t$): edge modes, protected degeneracy.
- Trivial ($|\mu| > 2t$): no edge modes.

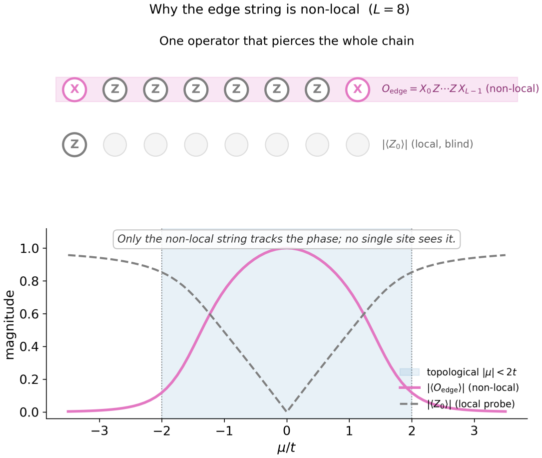
A **bulk invariant** (winding number) labels the phases; it is a property of the whole band, not of any single site.

Block 2 – Onto Qubits, and a Non-Local Diagnostic

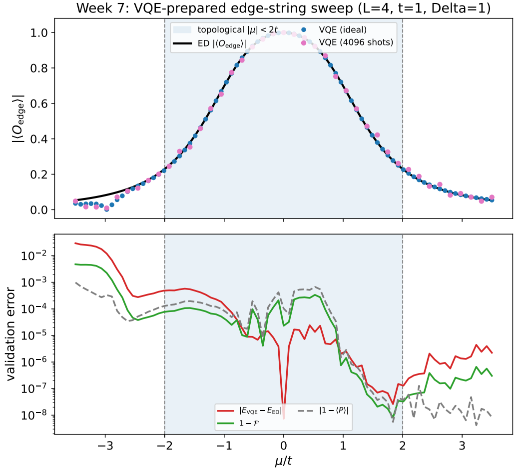
Jordan–Wigner maps the fermion chain to L qubits. Two objects survive the mapping as the physics:

- **Fermion parity** $\hat{P} = \prod_j Z_j$, a conserved $\mathbb{Z}_2$ charge ($[H, \hat{P}] = 0$).
- The **edge string** $X_0 Z_1 \cdots Z_{L-2} X_{L-1}$ – the operator that links the two ends.

No *local* operator (e.g. $\langle Z_0 \rangle$) can see the topological order; only the string does.



Block 3 – Preparing the State with a VQE



A parity-constrained **EfficientSU2** ansatz is optimized across μ (warm-started to stay in the even sector). The VQE-prepared state reproduces the exact edge-string diagnostic: high inside the topological window, vanishing outside.

This is the *noiseless* circuit we now expose to a device.

Block 4 – The NISQ Reality Check

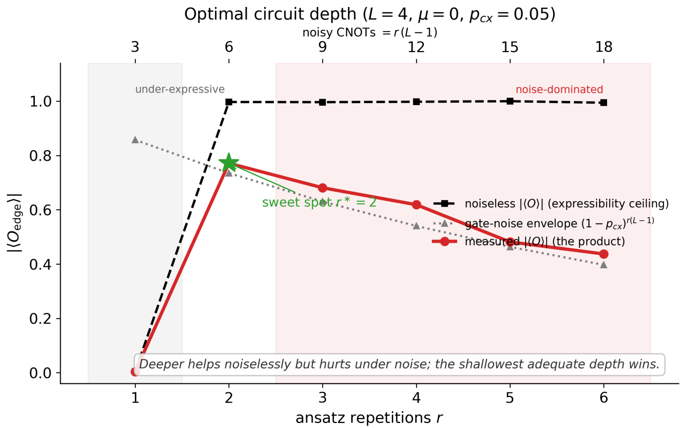
We keep the *same* validated VQE state and evolve it as a **density matrix** through a Qiskit `NoiseModel`: a quantum channel fires after every gate, and the diagnostic is read as $\mathrm{Tr}[X_0 Z_1 \cdots Z_{L-2} X_{L-1}\, \rho]$.

- **Depolarizing / thermal (T_1, T_2) / phase / amplitude** damping, applied at the circuit level.
- Verified to machine precision ($\sim 10^{-15}$) against an independent gate-by-gate reference evolution.

Four questions Block 4 answers

How deep should the circuit be? Which errors are recoverable? Is parity "topologically" protected? Does a longer chain help?

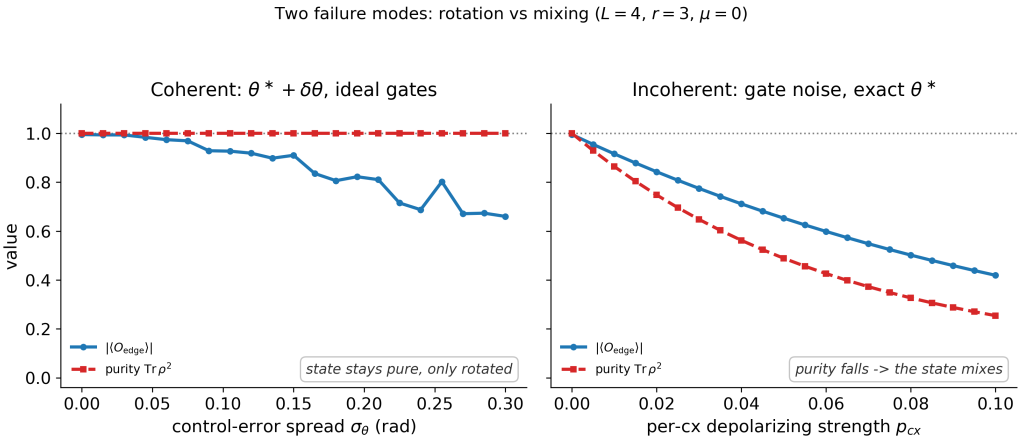
Block 4 – The Depth Optimum



Expressibility is a **threshold**: below a minimum depth the ansatz cannot represent the state; above it, the ideal signal saturates. Gate noise then *decays* as $(1-p_{cx})^{r(L-1)}$.

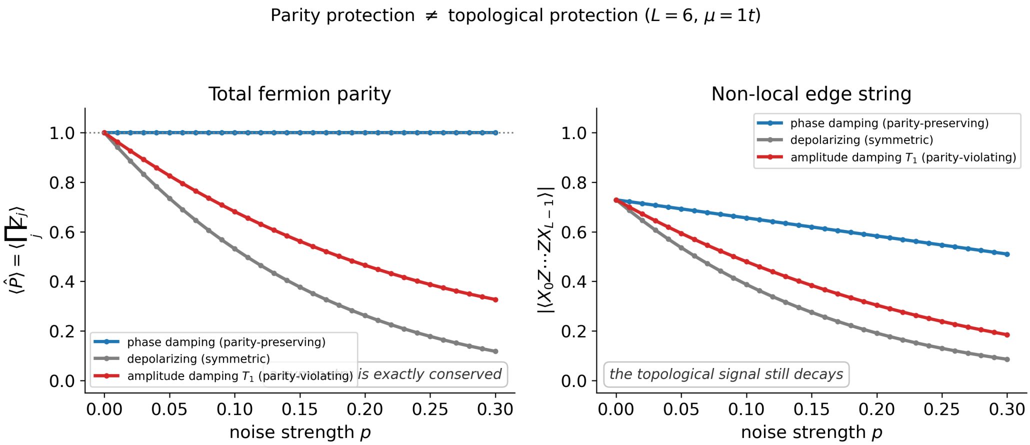
The optimum is the **shallowest adequately-expressive depth** – just deep enough, never deeper.

Block 4 – Two Families of Error



Coherent (miscalibrated angles $\theta^* + \delta\theta$, ideal gates): the state stays *pure* ($\mathrm{Tr}\,\rho^2 = 1$) – recoverable by recalibration. **Incoherent** (depolarizing channel): genuine decoherence – signal *and* purity fall, and the loss cannot be undone.

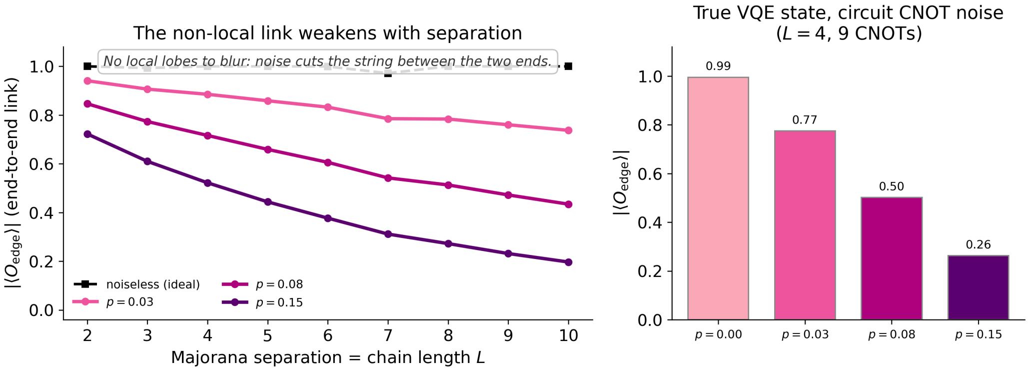
Block 4 – Parity Protection $\neq$ Topological Protection



Left: parity-preserving phase damping keeps $\langle \hat{P} \rangle = 1$ exactly; parity is a *symmetry* ($\mathbb{Z}_2$), protected in *every* phase. **Right:** the topological edge string still decays under every channel – the symmetry is not the topology.

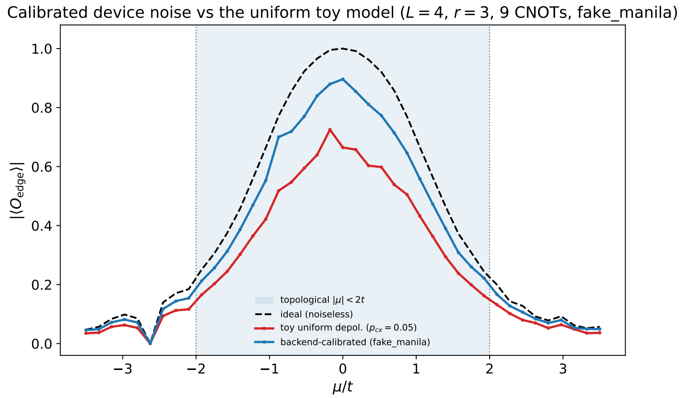
Block 4 – Noise Severs the Non-Local Link

H1 under noise: in the qubit/VQE encoding the Majorana signal is non-local, so noise severs it



The Majorana signal is carried by the non-local string, so there are *no* local lobes to hide behind: noise cuts the link between the two ends. **Left:** the link weakens with separation L. **Right:** on the true VQE state, circuit CNOT noise drives $|\langle X_0 Z_1 \cdots Z_{L-2} X_{L-1}\rangle|$ from 0.99 down to 0.26.

Block 4 – From a Toy Model to a Real Device



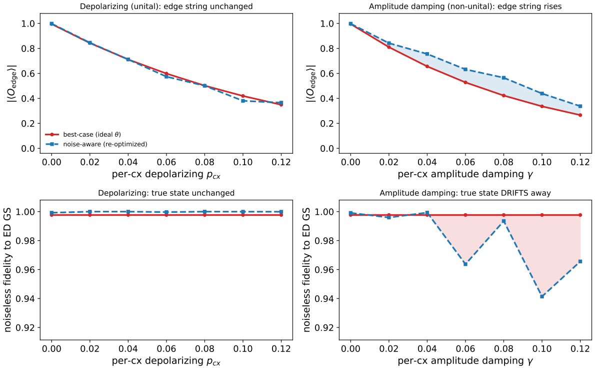
Replacing the hand-set uniform noise with `NoiseModel.from_backend` pulls in a real device's recorded calibration (IBM `fake_manila`: per-qubit T_1/T_2, per-gate error rates ~ 0.01).

- At $\mu = 0$: ideal 0.99 $\to$ **device** 0.87 $\to$ uniform toy ($p = 0.05$) 0.67.
- The calibrated curve sits *between* the two: a uniform toy model **overstates** the damage.

Caveat: the density matrix is read directly, so backend *readout* error is excluded – even the device curve is mildly optimistic.

Block 4 – Optimizing the Noisy Cost Is Not Preparing the State

Optimizing the noisy cost is not preparing the state: non-unital noise inflates the edge string while the true-state fidelity falls ($L = 4$, $r = 3$, $\mu = 0$, $\lambda = 0.1$)



The noise now lives *inside* the optimizer – every cost evaluation is a noisy density-matrix run.

- **Unital** (depolarizing): noise-aware = best-case. This *validates* the noiseless-preparation idealization used throughout.
- **Non-unital** (amplitude damping): the optimizer **raises** the edge string (top) while true-state fidelity **falls** (bottom).

A better *number*, a worse *state*.

Synthesis – What the Project Establishes

1. **The topology is real, but the signal is non-local.** A single edge-string observable tracks the phase from exact diagonalization all the way to a noisy circuit – no local probe can.
2. **Parity is symmetry-protected, not topologically protected.** It can be exactly conserved under noise while the topological signal decays: the two must not be conflated.
3. **On NISQ, depth is the lever.** Signal $=$ expressibility threshold $\times$ accumulated gate noise; the optimum is the shallowest adequately-expressive circuit.
4. **Optimizing the device cost $\neq$ preparing the state.** Under non-unital noise a noise-aware VQE can inflate the measured edge string while its fidelity to the true ground state falls – so we report the honest number, not the flattering one.

A longer chain sharpens the physics (smaller parity gap) but costs more gates – the device and the physics pull in opposite directions.

What We Built

Physics engine

- Exact BdG diagonalization, winding invariant, finite-size scaling.
- Jordan–Wigner qubit Hamiltonian and parity sectors.

Quantum circuits

- Parity-constrained VQE (Qiskit), warm-started sweeps.

Noise laboratory

- Circuit-level `NoiseModel` (density-matrix), machine-precision verified.
- Depth, channel, coherent/incoherent, and length studies.

Communication

- Intuition-first figure gallery, published to GitHub Pages.

Outlook

- **Error mitigation.** Zero-noise extrapolation / readout correction on the edge-string estimator – can the non-local signal be recovered post hoc?
- **Scaling to larger L.** The dense density-matrix path walls near $L \approx 15$; a sparse even-sector solver with statevector VQE and an *ED-free* selection criterion would reach $L \sim 20$–24 on HPC.
- **Beyond the minimal model.** Longer chains, disorder, and interactions toward a more realistic topological wire.

Two prior outlook items are now *done* (this deck): a `NoiseModel`-backed **noisy VQE optimization** and **backend-calibrated** noise from real device data.

Thank You

Majorana Modes in the Machine

From the Kitaev chain to a noisy processor – one non-local signal, followed end to end.

Code: `src/` (blocks 1–4, `showcase.py`). Notes: `notes/`. Figures: GitHub Pages gallery.